

Executive Insight Summary

Nigeria Fuel Price Trend & Volatility Analysis (2007–2026)

Fellow: Joseph Eduoku

Fellow ID: FE/26/72213414

Track: Data Science NextGen Cohort

ALC: Guru Innovation Hub, Cross River State

Tools: Microsoft Excel, Power Query, Power BI, and DAX

1. Project Overview

This project analyses Nigeria's Premium Motor Spirit (PMS) retail price data from 2007 to 2026 to understand how fuel prices have changed over time and how pricing differs across states and geopolitical zones. Particular attention is given to the significant price shift following the removal of the fuel subsidy in May 2023.

2. Key Findings

- **Current National Average:** ₦1,097.21 per litre
- **Highest Recorded Price:** ₦1,601.34 per litre
- **Average Historical Year-on-Year Change:** 17.18%
- **Subsidy-Removal Price Impact:** 1.29× immediate increase indicator

The analysis shows a clear change in Nigeria's fuel-price pattern after May 2023. Before the subsidy removal, petrol prices generally remained below ₦250 per litre for much of the historical period. Following the policy change, prices increased substantially, reaching more than ₦1,000 per litre in the later period of the dataset.

3. Regional Price Differences

Fuel-price increases were not uniform across Nigeria. The **North-East** recorded the highest post-subsidy average at approximately **₦736 per litre**, compared with about **₦131 per litre** before the subsidy removal. The **North-West** followed at approximately **₦588 per litre**, compared with **₦123 per litre** before the policy change.

The **South-West** and **South-East** recorded lower post-subsidy averages of approximately **₦549** and **₦519 per litre**, respectively. These differences indicate that location, transportation costs, supply routes, and access to major fuel distribution points can influence retail fuel prices.

4. State-Level Insights

The historical analysis identifies several northern states among those with the highest long-term average prices, including:

1. **Borno — ₦243/L**
2. **Yobe — ₦222/L**
3. **Adamawa — ₦214/L**
4. **Sokoto — ₦210/L**

These differences highlight the potential effect of distribution distance, transportation costs, and regional supply conditions on fuel prices.

5. Recommendations

1. **Improve fuel distribution infrastructure:** Strengthen transportation and distribution networks to reduce regional price differences.
2. **Develop regional storage capacity:** Establish strategic fuel storage facilities in areas vulnerable to supply disruptions and price volatility.
3. **Expand alternative energy options:** Increase investment in CNG and other alternative energy solutions, particularly for public transportation.
4. **Monitor regional price disparities:** Use data-driven monitoring systems to identify unusual price increases and improve policy responses.

Conclusion

The analysis demonstrates a major transformation in Nigeria's petrol pricing landscape between 2007 and 2026. The removal of the fuel subsidy marked a significant shift toward higher and more variable retail prices, while substantial differences remain across regions and states. Data-driven monitoring, improved distribution infrastructure and alternative energy investments can help reduce the impact of future fuel-price shocks on households and businesses.